

Assembly Symmetry

- [How to use it?](#)
- [Particular BL:](#)
- [ADS Naming rules for duplicated objects:](#)
 - [KIS Symmetry improvements \(N2H-29303\)](#)
- [Properties:](#)
 - [Refer To Symmetry improvements \(N2H-29301\)](#)
- [As Is](#)
- [To BE:](#)
- [Video to be added after testing for story N2H-29301](#)
- [Update attributes transfer when creating a symmetric object \(N2H-27374\)](#)
- [Technological Notes](#)
- [Update Instance Attributes after creating a handed element/assembly on Fasteners \(N2H-28227\)](#)

EIS	PI	Owner	Capability	Feature	Story
Calad an	26	Mavericks	T3DEXPOP-18774 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-20025 - Détails de la demande en cours d'obtention... ÉTAT	N2H-19730 - Détails de la demande en cours d'obtention... ÉTAT
Dago bah	27	Mavericks	T3DEXPOP-18774 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-20023 - Détails de la demande en cours d'obtention... ÉTAT	N2H-19727 - Détails de la demande en cours d'obtention... ÉTAT
Dago bah	27	Mavericks	T3DEXPOP-20072 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-21262 - Détails de la demande en cours d'obtention... ÉTAT	N2H-21568 - Détails de la demande en cours d'obtention... ÉTAT
Dago bah	27	Mavericks	T3DEXPOP-20072 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-21262 - Détails de la demande en cours d'obtention... ÉTAT	N2H-20888 - Détails de la demande en cours d'obtention... ÉTAT
Jakku	33	Mavericks	T3DEXPOP-27518 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-27518 - Détails de la demande en cours d'obtention... ÉTAT	N2H-27374 - Détails de la demande en cours d'obtention... ÉTAT
Krypton	34	Mavericks	T3DEXPOP-28087 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-27906 - Détails de la demande en cours d'obtention... ÉTAT	N2H-28110 - Détails de la demande en cours d'obtention... ÉTAT
Krypton	34	Mavericks	T3DEXPOP-28087 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-27906 - Détails de la demande en cours d'obtention... ÉTAT	N2H-28227 - Détails de la demande en cours d'obtention... ÉTAT
Krypton	34	Mavericks	T3DEXPOP-26887 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-26792 - Détails de la demande en cours d'obtention... ÉTAT	N2H-28726 - Détails de la demande en cours d'obtention... ÉTAT
Krypton	34	Mavericks	T3DEXPOP-25274 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-27905 - Détails de la demande en cours d'obtention... ÉTAT	N2H-28028 - Détails de la demande en cours d'obtention... ÉTAT
Lusitania	35	Mavericks	T3DEXPOP-29036 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-28932 - Détails de la demande en cours d'obtention... ÉTAT	N2H-29303 - Détails de la demande en cours d'obtention... ÉTAT

Lusitania	35	Mavericks	T3DEXPOP-29036 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-28932 - Détails de la demande en cours d'obtention... ÉTAT	N2H-29301 - Détails de la demande en cours d'obtention... ÉTAT
Lusitania	35	Mavericks	T3DEXPOP-29036 - Détails de la demande en cours d'obtention... ÉTAT	T3DEXPOP-28931 - Détails de la demande en cours d'obtention... ÉTAT	N2H-29340 - Détails de la demande en cours d'obtention... ÉTAT

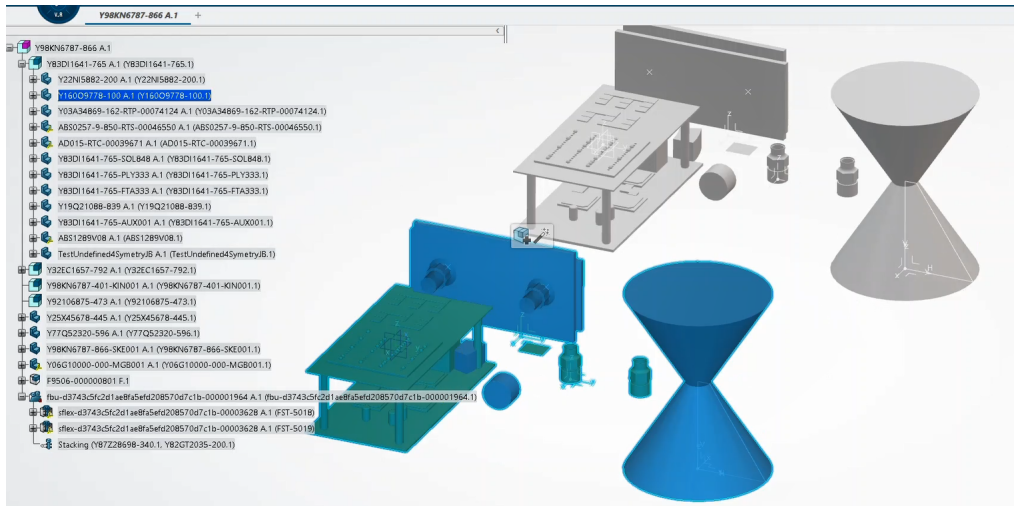


The "Assembly Symmetry" function can be used to duplicate an assembly.

This can be used on EM or SA.

How to use it?

- Click on "assembly symmetry"
- Select the assembly to duplicate in the stree structure
- a preview will be displayed
- validate
- the new assembly is opened in a new tab.



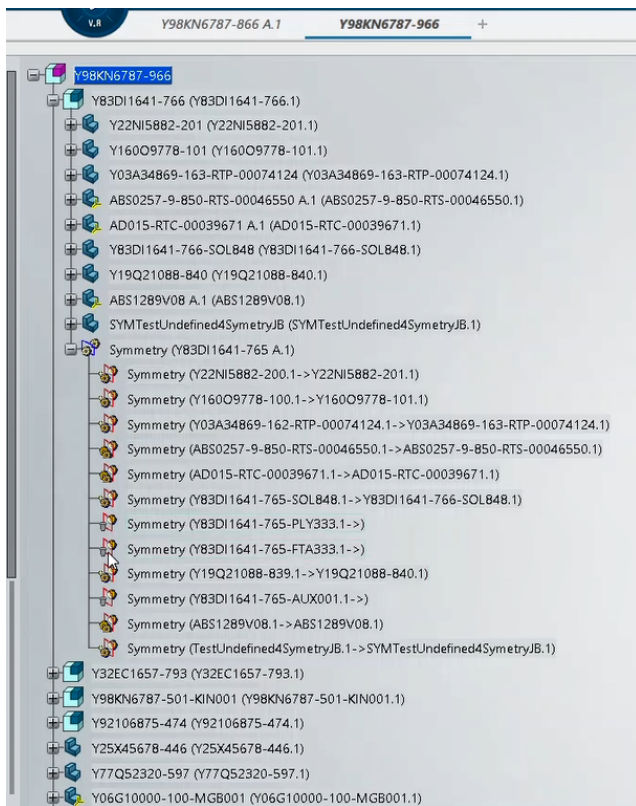
Example of preview.

Particular BL:

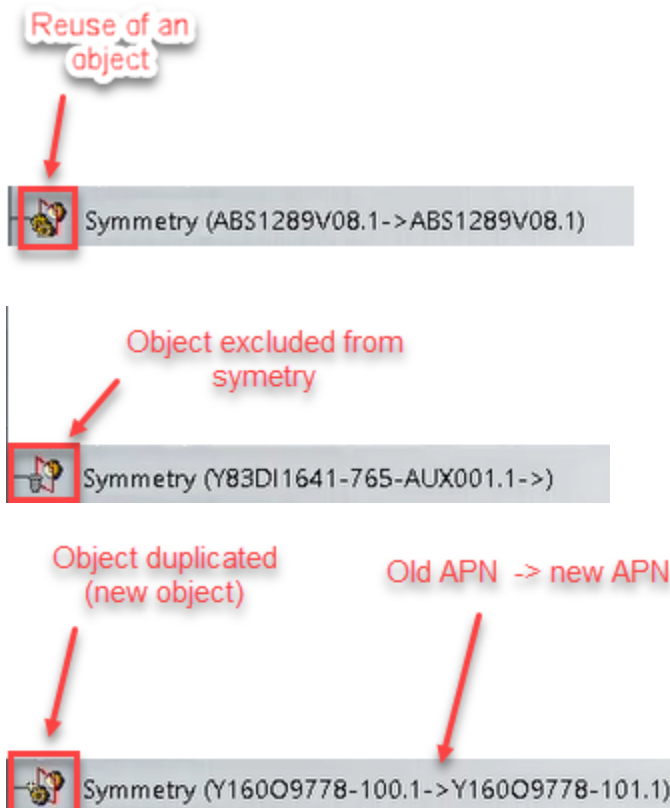
During usage of symmetrical function, some element can be handed, some reuse, some excluded automatically.

- for element selected to become a symmetrical item, naming rules shall be respected as defined in the linked file [naming rules for sym.xlsx](#)
- part to be exclude during symmetry function mentionned in the same file. Trigger is TCode and Business Type. if Tcode does not exist, business type only
- part to reuse automatically during symmetry function mentionned in the same file. Trigger is TCode and Business Type if Tcode does not exist, business type only

In the generated object, you will find information of each actions that have been done on each object:



You can find the following informations:



ADS Naming rules for duplicated objects:

List of reused object (APN Remain same as original one)

- Standart parts
- Refer to with tCode:
 - RTS
 - RTM
 - RTC
 - RTD
- 3DEQ

List of excluded object:

- Specialized items
- FTA
- Extended Item with tCode:
 - PLY
- Fasteners & bundle fasteners

List of duplicated object with APN Rules:

- EP, no matter tCode
 - Identical 13 digit + 1
 - ex: Y28220001-**001** Y28220001-**002**
- SA, no matter tCode
 - Identical 13 digit + 1
 - ex: Y28220001-**001** Y28220001-**002**
- Undefined
 - SYM prefix added
 - ex: Test **SYM**Test
- Refer TO:
 - RTA: Identical 13 digit + 1-RTA-BOM Reference
 - ex: Y50220001-**401**-RTA-12345678 Y50220001-**402**-RTA-12345678
 - RTP: Identical 13 digit + 1-RTP-BOM Reference
 - ex: Y50220001-**401**-RTP-12345678 Y50220001-**402**-RTP-12345678
 - RTH: Identical 13 digit + 1-RTH-BOM Reference
 - ex: Y50220001-**401**-RTH-12345678 Y50220001-**402**-RTH-12345678
- SAM, no matter tCode
 - Identical 13 digit + 1
 - ex: Y92S12345-**001** Y92S12345-**002**
- Master geometry, no matter tCode (Tcode KEY for example in APN here)
 - Identical 13 digit + 1 if SA symmetry
 - ex: Y06G12345-**001**-KEY001 Y06G12345-**002**-KEY001
 - Identical 11 digit + 1 if EM symmetry
 - ex: Y06G12345-**001**-KEY001 Y06G12345-**101**-KEY001
- Design Simulation, except KIS Tcode (Tcode DUP for example in APN here)
 - Identical 13 digit + 1 if SA symmetry
 - ex: Y53K30123-**401**-DUP001 Y53K30123-**402**-DUP001
 - Identical 11 digit + 1 if EM symmetry
 - ex: Y53K30123-**401**-DUP001 Y53K30123-**501**-DUP001
 - Tcode KIS under Design Simulation with Tcode KIN or Design Solution
 - Identical 9 digits of the original-counter +1-KISzzz with zzz from 001 to 999
 - Original APN : **Y53KQ2345-401-KIS001**
 - Handed Element APN : **Y53KQ2345-402-KIS001**
- EHI, no matter tCode
 - Identical 13 digit + 1
 - ex: Y28220001-**001** Y28220001-**002**
- Project Definition, no matter tCode
 - Identical 13 digit + 1
 - ex: Y28X20001-**401** Y28X20001-**402**

KIS Symmetry improvements ([N2H-29303](#))

As a Designer I want to be able to create a symmetry onto a -KIS design simulation and rename it automatically

As Is :

Fastener Bundle	Business Type	3DEQ Sub-Type	Technological Code (Tcode)	Numbering Original	Numbering Handed for implementation	Numbering rules due to ootb BL rules limitation	Numbering Handed Business Needs	Handed numbering example
EM	Design Simulation	No 3DEQ SubType	KIN	Y53K30123-401-KIN001 Y53K30123-801-KIN001	Identical 13 digit + 1 if SA symmetry 11 digit + 1 if EM symmetry	if digit 10 is different to 9 or 8 then 13 digit + 1 else 11 digit + 1	Identical 9 digit "-" counter + 1 if SA symmetry 9 digit "-" counter + 100 if EM symmetry	Y53K30123-402-KIN001 Y53K30123-901-KIN001
EM	Design Simulation	No 3DEQ SubType	DUP	Y53K30123-401-DUP001 Y53K30123-801-DUP001	Identical 13 digit + 1 if SA symmetry 11 digit + 1 if EM symmetry	if digit 10 is different to 9 or 8 then 13 digit + 1 else 11 digit + 1	Identical 9 digit "-" counter + 1 if SA symmetry 9 digit "-" counter + 100 if EM symmetry	Y53K30123-402-DUP001 Y53K30123-901-DUP001
EM	Design Simulation	No 3DEQ SubType	HUM	Y53H30345-401-HUM001 Y53H30345-801-HUM001	Identical 13 digit + 1 if SA symmetry 11 digit + 1 if EM symmetry	if digit 10 is different to 9 or 8 then 13 digit + 1 else 11 digit + 1	Identical 9 digit "-" counter + 1 if SA symmetry 9 digit "-" counter + 100 if EM symmetry	Y53H30345-402-HUM001 Y53H30345-901-HUM001

To Be

Fastener Bundle	Business Type	3DEQ Sub-Type	Technological Code (Tcode)	Numbering Original	Numbering Handed for implementation	Numbering rules due to ootb BL rules limitation	Numbering Handed Business Needs	Handed numbering example
EM	Design Simulation	No 3DEQ SubType	KIN	Y53K30123-401-KIN001 Y53K30123-801-KIN001	Identical 13 digit + 1 if SA symmetry 11 digit + 1 if EM symmetry	if digit 10 is different to 9 or 8 then 13 digit + 1 else 11 digit + 1	Identical 9 digit "-" counter + 1 if SA symmetry 9 digit "-" counter + 100 if EM symmetry	Y53K30123-402-KIN001 Y53K30123-901-KIN001
EM	Design Simulation	No 3DEQ SubType	DUP	Y53K30123-401-DUP001 Y53K30123-801-DUP001	Identical 13 digit + 1 if SA symmetry 11 digit + 1 if EM symmetry	if digit 10 is different to 9 or 8 then 13 digit + 1 else 11 digit + 1	Identical 9 digit "-" counter + 1 if SA symmetry 9 digit "-" counter + 100 if EM symmetry	Y53K30123-402-DUP001 Y53K30123-901-DUP001
EM	Design Simulation	No 3DEQ SubType	HUM	Y53H30345-401-HUM001 Y53H30345-801-HUM001	Identical 13 digit + 1 if SA symmetry 11 digit + 1 if EM symmetry	if digit 10 is different to 9 or 8 then 13 digit + 1 else 11 digit + 1	Identical 9 digit "-" counter + 1 if SA symmetry 9 digit "-" counter + 100 if EM symmetry	Y53H30345-402-HUM001 Y53H30345-901-HUM001
EM	Design Simulation	No 3DEQ SubType	KIS	Y53H30345-401-KIS001	Identical 13 digit + 1	13 digit + 1	Identical 9 digit "-" counter + 1	Y53H30345-402-KIS001

Properties:

During Edition of attribute (Edit properties, Spread sheet editor, Right-clc properties, and Sheet editor) on a symmetrical element, all attributes values shall be blocked for data with reference status In Work or Private, except this list of attributes:

- Airbus Part Number
- Description
- English Title



Refer To Symmetry improvements (N2H-29301)

The goal of this Feature is to be able to generate new Refer To Item generation during the symmetry creation without being blocked by the unicity Key.

Impacted items: VPM reference having the business Type set "Refer To" and the Techno Code set to :RTM,RTC, RTP, RTA and RTH

If during the symmetry generation or update the user select an item identified as Refer To (Techno Code **RTM,RTC, RTP, RTA and RTH**) the "Symmetric reference" option, then the new 'Refer To' generated needs to have its Airbus Part Number renamed by using the last 8 digits or the new PRD generated.

e.g if the Original APN is **AIMS12345-RTM-12345678** then the symmetric "Refer To" will be renamed (APN and Title) **AIMS12345-RTM-#####** with ##### the last 8 digits of the new PRD of the item created

As Is

Parent	Business Type	3DEQ Sub-Type	Technological Code (Tcode)	Numbering Original	Numbering Handed for implementation	Numbering rule
EM, Storable Assembly, Project Definition - IFA	Refer to	No 3DEQ SubType	RTP	Y28220001-001-RTP-12345678 BOM Reference=Y28220001-001	Identical 13 digit + 1 BOM Reference 13 digit + 1 Replace 8 last digits by the new 8 last digits of the prd naming	
EM, Storable Assembly, Project Definition - IFA	Refer to	No 3DEQ SubType	RTM	AIMS10-07-010-RTM-12345678	Replace 8 last digits by the new 8 last digits of the prd naming	
EM, Storable Assembly, Project Definition - IFA	Refer to	No 3DEQ SubType	RTA	Y50220001-401-RTA-12345678 BOM Reference=Y50220001-401-RTA-1234567	Identical 13 digit + 1 BOM Reference 13 digit + 1 Replace 8 last digits by the new 8 last digits of the prd naming	
EM, Storable Assembly, Project Definition - IFA	Refer to	No 3DEQ SubType	RTC	AA451-RTC-12345678	Replace 8 last digits by the new 8 last digits of the prd naming	
EM	Refer to	No 3DEQ SubType	RTH	Y92921234-401-RTH-12345678 BOM Reference=Y92921234-401	Identical 13 digit + 1 BOM Reference 13 digit + 1 Replace 8 last digits by the new 8 last digits of the prd naming	

To BE:

Parent	Business Type	3DEQ Sub-Type	Technological Code (Tcode)	Numbering Original	Numbering Handed for implementation
EM, Storable Assembly, Project Definition - IFA	Refer to	No 3DEQ SubType	RTP	Y28220001-001-RTP-12345678 BOM Reference=Y28220001-001	Identical 13 digit + 1 BOM Reference 13 digit + 1
EM, Storable Assembly, Project Definition - IFA	Refer to	No 3DEQ SubType	RTA	Y50220001-401-RTA-12345678 BOM Reference=Y50220001-401-RTA-1234567	Identical 13 digit + 1 BOM Reference 13 digit + 1
EM	Refer to	No 3DEQ SubType	RTH	Y92921234-401-RTH-12345678 BOM Reference=Y92921234-401	Identical 13 digit + 1 BOM Reference 13 digit + 1

Video to be added after testing for story N2H-29301

Update attributes transfer when creating a symmetric object ([N2H-27374](#))

- Allow the Inherit, Synchronisation and Manual modification of all the **References** attributes mentioned below for ALL the **Lower Level Entity objects where the Symmetry is allowed (3Dshape) --> (FTA, Dummy)**.
- Update the symmetry behavior according to the one on the following table:

Technological Notes

Techno Notes are set automatically by Default for handed Elements (Symmetrical parts) **for References Only not Instances**

Business Types having these attributes concerned by symmetry

- EP **Yes**
- SA **Yes**
- Refer to **Yes**
- Extended Item **Yes** (only for SOL T-Code)
- SAM **Yes**
- Master Geometry **Yes**
- Undefined **Yes**
- EH **Yes**

General Techno Note

"001-ED03-48 PART NOMINAL GEOMETRY AND CATIA FTA INFORMATION ON CAD FILE ## ISSUE ##"

Where the ## values are:

1st ## is the PN of the original Part

2nd ## is the ** version of the original part (where the symmetry link is pointing)

For example for PN Y53201000-002 A:

General Technological Notes: "001-ED03-48 PART NOMINAL GEOMETRY AND CATIA FTA INFORMATION ON CAD FILE Y53201000-001 ISSUE A"

Specific Technological Notes : Empty.

Attribute	Inherit (YES/NO)	Comments	Attribute Update (YES/NO)	Attribute Modifiable Manually	Valid For
Airbus Part Number	YES*	To be evolved according to GAP 8	NO	YES	All Lower Level Entities
Revision	NO		-	-	All Lower Level Entities
English Title	YES		YES	YES	All Lower Level Entities
Business Type	YES		YES	NO	All Lower Level Entities
Technological Code	YES		YES	NO	All Lower Level Entities (except Undefined)
Maturity State	NO		-	-	All Lower Level Entities
Reference Status	NO		-	-	All Lower Level Entities
Design Stage	YES		YES	YES	All Lower Level Entities (except Design Project Definition, Design Simulation)
Identifiable Part	YES		YES	YES	EP, SA, DS and EHI only
Functional Class	YES		YES	YES	EP, SA, DS, EHI
Marking = Labelled	YES		YES	YES	EP, SA, DS, EHI
Material	YES		YES	YES	EP editable (SA, DS, EHI present but not editable)
General Technological Notes	NO		NO	YES (through Know how apps)	All Lower Level Entities (except Design Project Definition, Design Simulation)
Specific Technological Notes	NO		NO	YES (through Know how apps)	All Lower Level Entities (except Design Project Definition, Design Simulation)
Protective Treatment Code	YES		YES	YES	EP, SA, DS
Surface Finish Reference	YES		YES	YES	EP, SA, DS
Limits Not Stated = General Tolerances	YES		YES	YES	EP, SA, DS
Description	YES		YES	YES	All Lower Level Entities
Manual Associated Documents	YES		YES	YES	All Lower Level Entities
Raw Material	YES		YES	YES (through Know how apps)	EP
Alternate Material	YES		YES	YES (through Know how apps)	EP editable (SA, DS present but not editable)
Material Quantity	YES*		YES	YES	EP, RTC and RTM
Roughness	YES		YES	YES	EP, SA and Design Solution

- To allow the Inherit, Synchronisation and Manual modification of all the **Instance** attributes mentioned below for ALL the **Lower Level Entity** objects where the **Symmetry is allowed**.
- To update the symmetry behavior according to the one on the following table:

Table Valid for all LLE instances (N2H-28110)

Instance Attribute	Inherit (YES/NO)	Comments	Attribute Update (YES/NO)	Attribute Modifiable Manually	Attributes Present on
Instance Design Stage	YES		YES	YES	All Lower Level Entities
General Technological Notes	YES		YES	YES (through Know apps)	All Lower Level Entities
Specific Technological Notes	YES		YES	YES (through Know apps)	All Lower Level Entities
Instance Alternate	NO		YES	YES (through Know apps)	EP, SA
Description	YES		YES	YES	All Lower Level Entities

E.g Video >> [N2H28110-EP symmetry Instance alternate only editable and updated, but not inherit.mp4](#)

Update Instance Attributes after creating a handed element/assembly on Fasteners (N2H-28227)

The goal is to update the behavior of the symmetry to ensure that all the important information is also present in the symmetry elements at instance level (For LLE and Flexible Fastener Objects).

Symmetry Attributes on Flexible Fasteners (Primary and secondary Fasteners):

(T3DEXPOP-26297 & T3DEXPOP-27905)

Table Valid for Primary Fasteners only: (Flexible Fasteners instances only):

Instance Attribute	Inherit (YES/NO)	Attribute Update (YES/NO)	Attribute Modifiable Manually
Installation orientation Section	YES	YES	YES
Installation Information Section	YES	YES	YES
Overcoating Information Section	YES	YES	YES
Drilling operation Section	YES	YES	
Bonding Information Section	YES	YES	YES
Cleaning/Stripping Information Section	YES	YES	YES
Torque Value Information Section	YES	YES	

Table Valid for both, Primary and Secondary Fastener's: (Flexible Fasteners and Anchor Nuts instances). *These new attributes (original and opposite handed) shall not be modifiable in the properties panels, only throught the 3DX fastening APS* (see T3DEXPOP-26297 and T3DEXPOP-25877)

We are not allowed to inherit values of the following Attributes for Torque and Drilling since they are OOTB Attributes, so this part will be taken out and a SR will be created for the following attributes:

Instance Attribute	Inherit (YES/NO)	Attribute Update (YES/NO)	Attribute Modifiable Manually
Entry Hole Depth	YES	YES	YES (through Fastener's App)
Entry Hole Countersunk Angle	YES	YES	YES (through Fastener's App)
Entry Hole Type	YES	YES	YES (through Fastener's App)
Entry Hole Counterbore Diameter	YES	YES	YES (through Fastener's App)
Tightness	YES	YES	YES (through Fastener's App)
Boring Code	YES	YES	YES (through Fastener's App)
Drilling Diameters Nominal	YES	YES	YES (through Fastener's App)
Drilling Diameters Delta Min	YES	YES	YES (through Fastener's App)
Drilling Diameters Delta Max	YES	YES	YES (through Fastener's App)
Torque Code	YES	YES	YES (through Fastener's App)
Torque Nominal	YES	YES	YES (through Fastener's App)
Torque Delta Min	YES	YES	YES (through Fastener's App)
Torque Delta Max	YES	YES	YES (through Fastener's App)

Table Valid for Secondary Fastener's only: (Anchor Nuts instances)

Instance Attribute	Inherit (YES/NO)	Attribute Update (YES/NO)	Attribute Modifiable Manually
Installation orientation Section	YES	NO	NO
Installation Information Section	YES	NO	NO
Overcoating Information Section	YES	NO	NO
Bonding Information Section	YES	NO	NO
Cleaning/Stripping Information Section	YES	NO	NO

The secondary fastener can be modified only with the OoTB tool and will not impact the customized attributes

- Currently, is possible to update the instance attributes of fasteners via "instance attribute propagation" command, but also is needed via manual modification, due the fact that the all attributes are grey-out.

Secondary fastener Symmetry

The secondary fastener take into account during symmetry.

The instance attributes of the symmetric secondary fastener shall not be editable as the original ones.